

CBSE Practice Worksheet

Class 7 | Science | Difficulty: Hard | Set C

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the escape velocity from Earth's surface?
(A) 8 km/s
(B) 11.2 km/s
(C) 15 km/s
(D) 25 km/s
2. What is the magnetic quantum number for 3d orbital?
(A) -2 to +2
(B) -3 to +3
(C) -1 to +1
(D) 0 only
3. Which law relates induced EMF to rate of change of magnetic flux?
(A) Lenz's law
(B) Faraday's law
(C) Ampere's law
(D) Ohm's law
4. Which enzyme is responsible for DNA replication?
(A) DNA ligase
(B) DNA polymerase
(C) Helicase
(D) Primase
5. What is the coordination number in FCC structure?
(A) 6
(B) 8
(C) 12
(D) 14
6. Which reaction is endothermic?
(A) Combustion
(B) Neutralization
(C) Photosynthesis
(D) Respiration
7. What is the wavelength of Lyman series limit?
(A) 91.2 nm
(B) 121.6 nm
(C) 656.3 nm
(D) 364.6 nm
8. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%

(D) 75%

9. What is the hybridization of carbon in methane?

- (A) sp
- (B) sp^2
- (C) sp^3
- (D) sp^3d

10. Which type of mutation involves change in single nucleotide?

- (A) Deletion
- (B) Insertion
- (C) Point mutation
- (D) Frameshift

Section B: Fill in the Blanks (1 mark each)

- 11. The process of splitting water during photosynthesis is called ____.
- 12. The band gap in semiconductors is typically ____ eV.
- 13. The IUPAC name of CH_3CH_2CHO is ____.
- 14. The enzyme that converts prothrombin to thrombin is ____.
- 15. Raoult's law applies to ____ solutions.
- 16. The Krebs cycle occurs in the ____ of mitochondria.
- 17. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.
- 18. The uncertainty principle was given by ____.

Section C: Short Answer Questions (2 marks each)

- 19. Derive the equation for kinetic energy of gases from kinetic theory.
- 20. Explain Le Chatelier's principle with examples.
- 21. Describe the light-dependent and light-independent reactions of photosynthesis.
- 22. Explain hybridization with examples of sp, sp^2 , and sp^3 .
- 23. Explain the structure and working of a nuclear reactor.
- 24. Explain the mechanism of DNA replication.
- 25. Describe the mechanism of nerve impulse transmission.

--- End of Worksheet ---

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